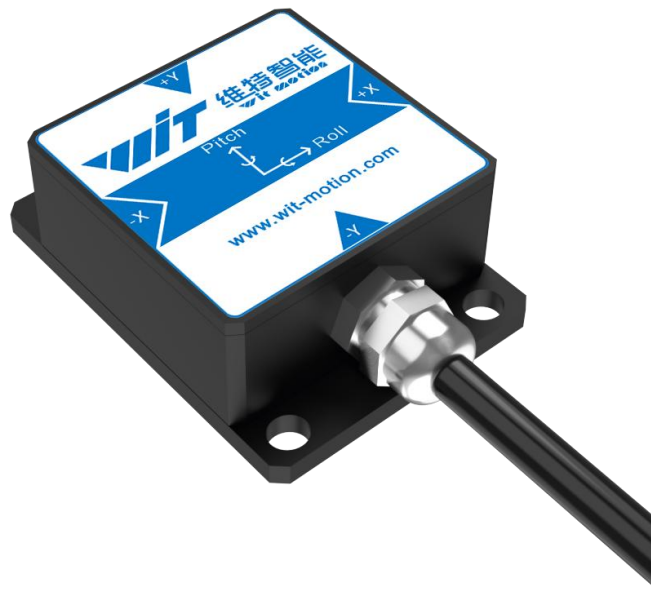




HWT901B Data Sheet & User Manual



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1. Product Introduction

1.1. Product Overview

- This product is a high-performance 3D motion attitude measurement system based on MEMS technology. It includes motion sensors such as a three-axis gyroscope , a three-axis accelerometer , a three-axis electronic compass , and a barometric pressure sensor . By integrating various high-performance sensors and utilizing a proprietary core attitude dynamics algorithm engine, combined with a high-dynamic Kalman filter fusion algorithm, it provides customers with high-precision, high-dynamic, and real-time compensation of three-axis attitude angles. Flexible configuration of various data types allows for diverse application scenarios.
- Leading sensor fusion algorithm based on Kalman filtering principle and with independent intellectual property rights, thus meeting various high-precision application requirements and achieving accurate motion capture and posture estimation.
- We have the domestically leading high-precision turntable equipment and instruments, and our products integrate independently developed high-precision calibration and calibration algorithms to improve the measurement accuracy of the products.
- At the same time, we provide users with various PC softwares, operating instructions, development manuals, and development codes required to minimize the R&D time for various needs.

1.2. Features

- The module integrates high-precision gyroscopes, accelerometers, and geomagnetic field sensors with magnetic recovery functions. It uses a high-performance microprocessor and advanced dynamics solver and Kalman dynamic filtering algorithms to quickly solve the module's current real-time motion posture.
- The use of advanced digital filtering technology can effectively reduce measurement noise and improve measurement accuracy.
- The module integrates an attitude solver, which, combined with the dynamic Kalman filter algorithm, can accurately output the current attitude of the module in a dynamic environment. The static attitude measurement accuracy is 0.1° , with extremely high stability, and its performance is even better than some professional inclinometers.
- The Z-axis heading angle incorporates geomagnetic sensor filtering and fusion, resolving the cumulative error caused by gyroscope integral drift in the 6-axis algorithm. This ensures long-term stable heading angle output.
Note: Due to magnetic field detection, calibration is required before use. During use, the device must be kept at least 20 cm away from

magnetic interference areas, electronic devices, magnets, speakers, and other hard magnetic objects.

- Military-grade magnetic field chip, automatically recovers from magnetic field interference.
- Data output content: time, acceleration, angular velocity, angle, magnetic field.
- The module has its own voltage stabilization circuit, operates in a 9~36V system, and is easy to connect.
- Dimensions: 55mm x 36.8mm x 24mm.
- Supports serial port, allowing users to choose the best connection method. The serial port rate is adjustable from 4800bps to 230400bps.
- P67 level protection, aluminum shell, strong resistance to external electromagnetic interference, stable performance, can adapt to long-term work in harsh industrial environments.

2. Parameter indicators

2.1. Accelerometer parameters

Parameter	Condition	Typical values
Range		±16g
Resolution	±16g	0.0005(g/LSB)
RMS noise	Bandwidth = 100 Hz	0.75~1mg-rms
Static zero drift	Horizontal placement	±20~40mg
Temperature Drift	-40°C ~ +85°C	±0.15mg/°C
Bandwidth		5~256Hz
Zero bias stability	10S smoothing	XY:15ug; Z:35ug
Zero bias instability	Allan variance	XY:10ug; Z:30ug

2.2. Gyroscope parameters

Parameter	Condition	Typical values
Range		$\pm 2000^{\circ}/s$
Resolution	$\pm 2000^{\circ}/s$	$0.061(^{\circ}/s)/(\text{LSB})$
RMS noise	Bandwidth = 100 Hz	$0.028 \sim 0.07(^{\circ}/s)\text{-rms}$
Static zero drift	Horizontal placement	$\pm 0.5 \sim 1^{\circ}/s$
Temperature Drift	$-40^{\circ}\text{C} \sim +85^{\circ}\text{C}$	$\pm 0.005 \sim 0.015 (^{\circ}/s)/^{\circ}\text{C}$
Bandwidth		$5 \sim 256\text{Hz}$
Zero bias stability	10S smoothing	$8^{\circ} /h$
Zero bias instability	Allan variance	$5^{\circ} /h$

2.3. Magnetometer parameters

Parameter	Condition	Typical values
Measuring range	Loop count value (200)	$-400\mu\text{T}$ to $+400\mu\text{T}$
noise	Loop count value (200)	20nT
Resolution @3V (LSB/ μT)	Loop count value (200)	38nT
Linearity $\pm 200\mu\text{T}$	Loop count value (200)	0.5%
Sensitivity	Loop count value (200)	26nT

2.4. Pitch and roll angle parameters

Parameter	Condition	Typical values
Range		X: $\pm 180^{\circ}$
		Y: $\pm 90^{\circ}$

Inclination accuracy		0.1°
Resolution	Horizontal placement	0.0055°
Temperature Drift	-40°C ~ +85°C	±0.5~1°

2.5. Heading angle parameters

Parameter	Condition	Typical values
Range		Z:±180°
Heading accuracy	9-axis algorithm, magnetic field calibration, dynamic/static	1° (without magnetic field interference) 【1】
	6-axis algorithm, static	0.5° (dynamic integral cumulative error) 【2】
Resolution	Horizontal placement	0.0055°

Note:

【1】 Please calibrate the magnetic field in the test environment before use to ensure that the sensor is familiar with the magnetic field in the environment. Please stay away from magnetic interference during calibration.

【2】 In some vibration environments, there will be cumulative errors. The specific error cannot be estimated. Please refer to the actual test.

2.6. Module parameters

2.6.1. Basic parameters

Parameter	Condition	Minimum	default	Maximum
Communication interface	485	4800bps	9600bps	230400bps
Output content	On-chip time, 3-axis acceleration, 3-axis angular velocity, 3-axis magnetic field, 3-axis angle, quaternion			

Output rate 【3】		0.2Hz	10Hz	200Hz
Startup time				1000ms
Operating temperature		-40℃		85℃
Storage temperature		-40 ℃		100 ℃
impact resistant				20000g

Note:

【3】 485 products use the Modbus protocol and are in master-slave mode, which is determined by the user's read content quantity and reading rate.

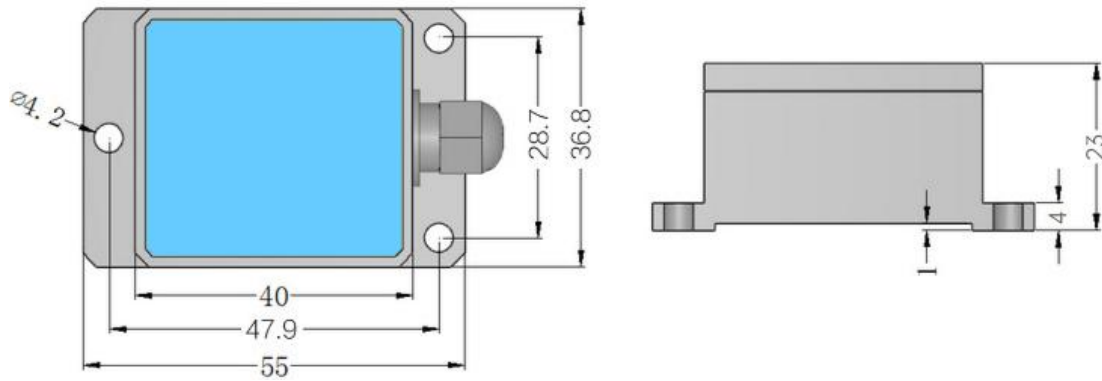
2.6.2. Electrical parameters

Note:

If you need 5V, please make a note to the salesperson when placing an order. Customization is supported.

parameter	condition	Minimum	default	Maximum
Supply voltage		9V	12V	36V
Working current	Working (12V)		4.6mA (TTL) 8.9mA (232) 8.5mA (485) 21.3mA (CAN)	

3. Product size



mm

4. Sensor connection

4.1. Line sequence definition and connection

TTL PIN Definition

Wiring	RED	YELLOW	GREEN	BLACK
	VCC 5-36V	TX	RX	GND



232 PIN Definition

Wiring	RED	YELLOW	GREEN	BLACK
	VCC 5-36V	232TX	232RX	GND



4.2. HWT901B connects to PC software

4.2.1. PC software download

1. CH340 driver download link:

https://drive.google.com/file/d/1JidopB42R9EsCzMAYC3Ya9eJ8JbHapRF/view?usp=drive_link

2. Witmotion software download link:

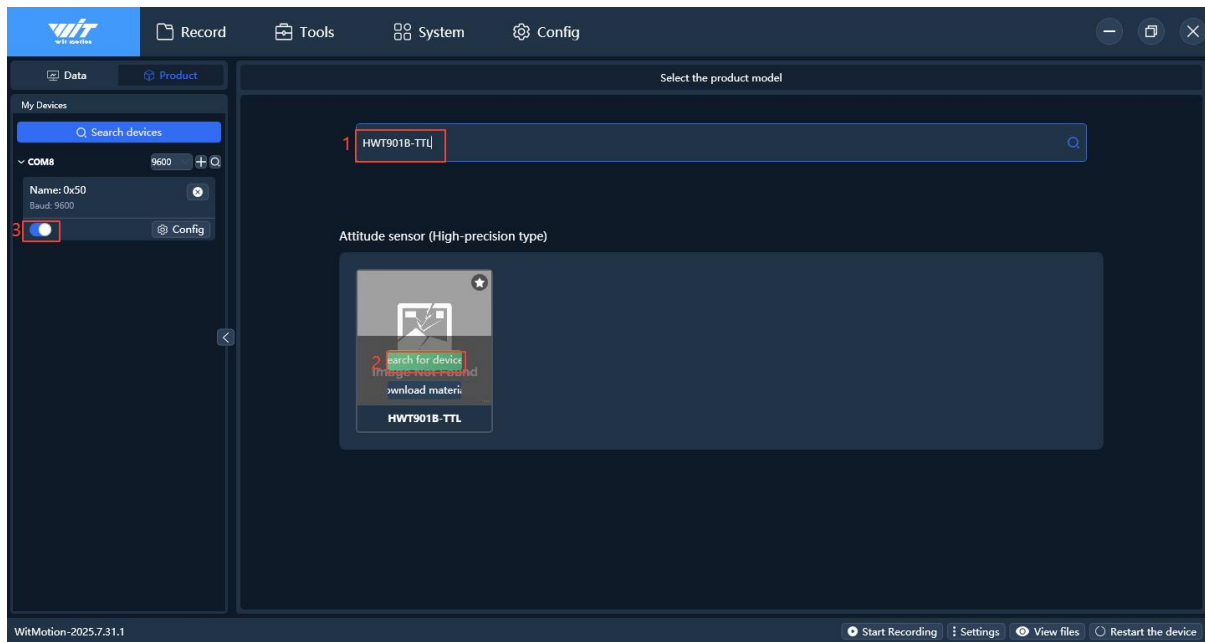
https://drive.google.com/file/d/10xysnkuyUwi3AK_t3965SLr5Yt6YKE-u/view?usp=drive_link

4.2.2. PC software connection method

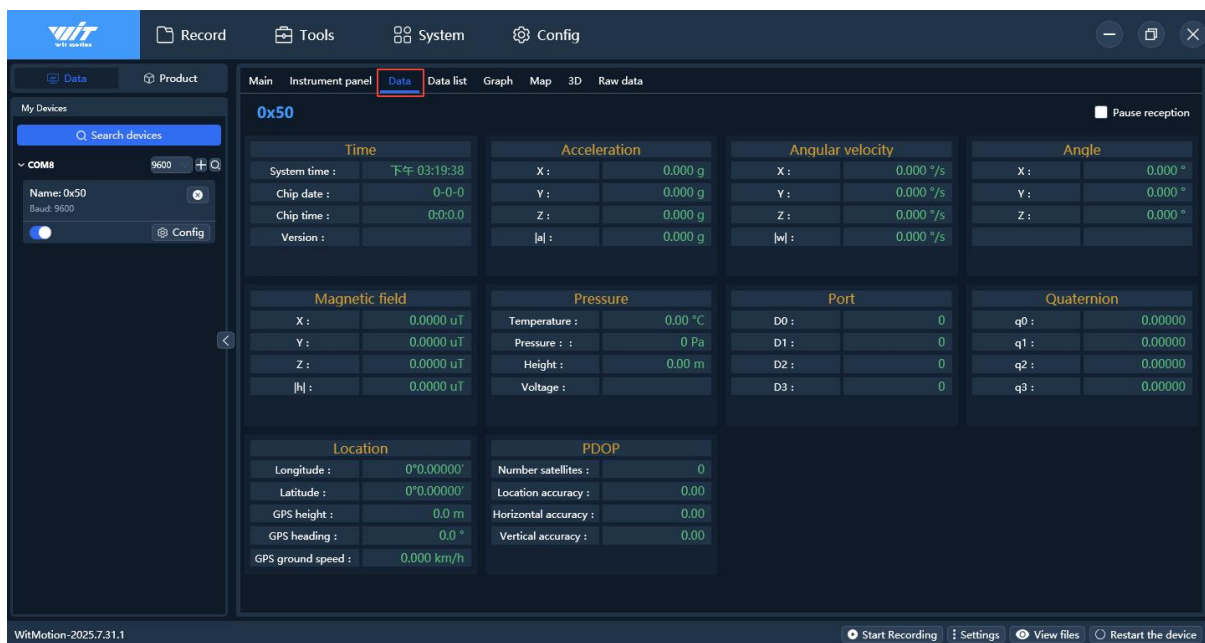
1. Open Witmotion software:

①Select model HWT901B-TTL or HWT901B-232;

②Click Search for device;



③Switch data interface;





5. Product Packaging

5.1. Sample packaging

